

Tracheostomy Management Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	Intensive Care Unit, Medical and surgical areas.	Medical, Nursing and Allied Health

2. Purpose

This document provides a procedural framework for staff in the management of tracheostomies and laryngectomies.

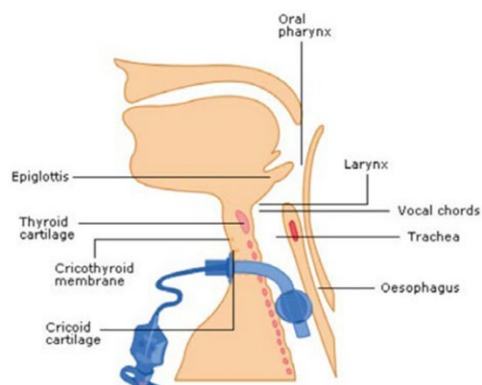
- For an Emergency: Call 55 or Activate Code Blue.
- For advice: Call ICU 2784.
- Refer to [Appendix 1](#): Emergency Tracheostomy Management Flowchart
- Do not cap a cuffed, inflated, non-fenestrated tube.
- Only a nurse experienced in tracheostomy tube management may escort a patient with a tracheostomy when not accompanied by a medical officer (MO).
- It is the responsibility of the surgeon or team who performed the initial insertion procedure to conduct the first tracheostomy tube change.
- The number of catheter passes when suctioning via a tracheostomy should be restricted to three times per suction episode and be no longer than 15 seconds each suction.
- The decision to commence oral diet and fluids is a medical decision.

3. Procedure

3.1 General information

The anterior incision is made below the cricoid cartilage through the 2nd – 4th tracheal rings (refer to figure 1 below).

Figure 1: Anatomy and position of tracheostomy tube



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When caring for a patient with a tracheostomy the nurse must be aware of the reason for the tracheostomy, the type of tube in situ and how it functions.

3.2 Indications for Tracheostomy

- To facilitate weaning from mechanical ventilation.
- Protect the airway.
- Angioedema.
- Maxillofacial trauma / head and neck surgery.
- Reduce anatomical dead space.
- Avoid a laryngeal injury.
- Aid in the management of tracheobronchial and pulmonary secretions.
- Upper airway obstruction.

4. Types of tracheostomy

Permanent: the creation of a tracheostomy following a total laryngectomy. The larynx is removed, and the trachea is sutured in position to form a permanent structure, known as a laryngectomy stoma.

Temporary: Is performed for patients as an elective procedure, e.g. at the time of major surgery such as total glossectomy.

Emergency: a tracheostomy may be performed as an emergency procedure when a patient has an obstructed airway.

- Among the more common conditions causing obstruction are trauma to the airway or neck, poisoning and infections.
- An emergency cricothyroidotomy may be indicated.

Surgical Tracheostomy: involves placement of a tracheostomy tube into the trachea by dissection and incision of the trachea under direct vision.

- This is performed in the Operating Theatre.

Percutaneous Dilatational Tracheostomy (PDT): is a bedside procedure performed using a needle and guide wire technique and dilatation of the trachea between the tracheal rings to facilitate placement of a tracheostomy tube.

Mini Tracheostomy: this method is used only when suctioning is required to remove retained tracheobronchial secretions.

- The procedure is also referred to as a cricothyroidotomy.
- The cricothyroid ligament is incised enabling a small endotracheal tube to mini- tracheostomy tube to be inserted.
- The mini tracheostomy has a small internal diameter which is often 4mm.

5. Tracheostomy site and tube

Figure 2: Tracheostomy site anatomy

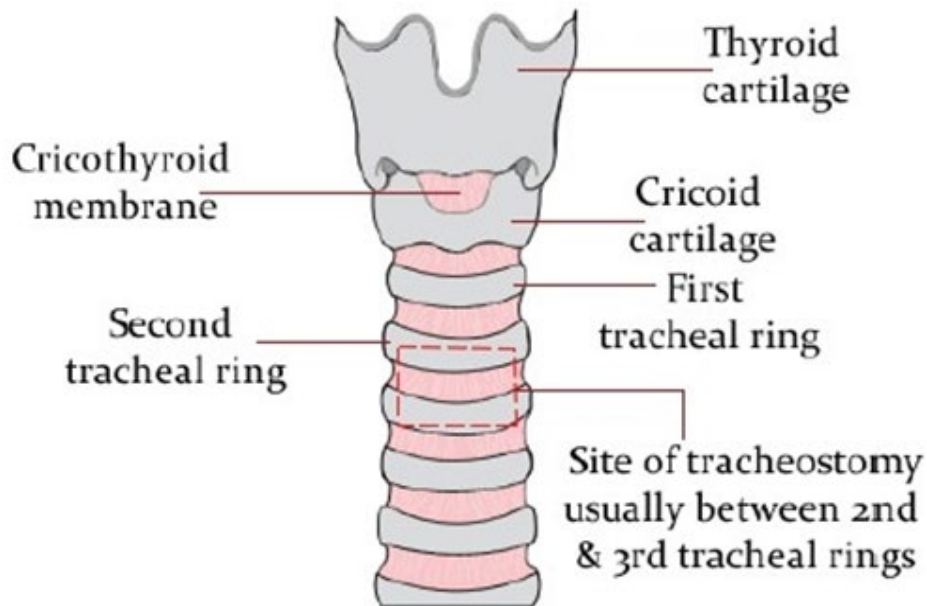
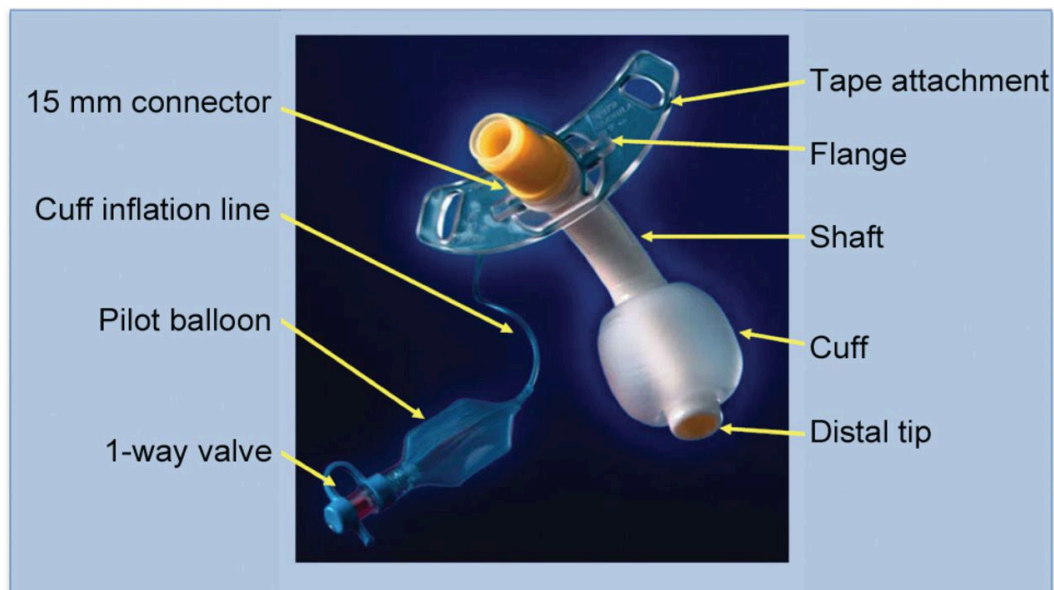


Figure 3: Components of tracheostomy tube



Note: Not all components are included on every tube, some are optional or manufacturer specific.

5.1 Types of tracheostomy tubes

- Tracheostomy tubes are available in different designs.
- Some have both an outer and inner cannula.
- The outer tube maintains the patency of the airway.
- The inner cannula fits firmly inside the outer tube and can be easily removed for cleaning without disturbing the stoma site. The following are some of the common tracheostomy tubes used:

Tube type	Specifics	Use	Example
Portex® cuffed and non - cuffed	Available with an inner cannula and also with an optional Suctionaid® port	Usually used for short term use as does not automatically have an inner cannula. Suctionaid® can be used to remove secretions that sit above the cuff prior to cuff deflation. Use cuff manometer to check cuff pressure.	 Figure 3: Cuffed tube with inner cannula
Shiley™ cuffed and non - cuffed	Available with or without fenestration. Plastic tube with an introducer and one inner cannula.	Use cuff manometer to check cuff pressure. Non - Cuffed tracheostomy tube may be used to maintain a patent airway especially when protection from secretions is not required or seal for ventilation.	 Figure 4: Fenestrated cuffed tracheostomy tube 6.
Mini-Trache®	Fine tube used when a person requires suctioning but is able to maintain and protect their airway. They do not have a very strong cough.	Suctioning only. Change 4 weekly Use cuff manometer to check cuff pressure.	 Figure 6: Mini trache kit 7
UniPerc™	Has an adjustable flange and reusable inner cannula	For patients with larger necks, muscular necks, short necks, obese necks and bull necks Use cuff manometer to check cuff pressure.	 Figure 7: UniPerc 8

6. Insertion of a Percutaneous Dilatational Tracheostomy Tube

6.1 General information

- May be performed in the Intensive Care Unit.
 - Percutaneous Dilatational Tracheostomy Tube (PDT) must be performed only on a stable patient by appropriately skilled staff in a controlled environment.
- It is contraindicated for emergency access.

6.2 Equipment

- Sedating and paralysing agents as prescribed by the MO on the medication chart.
- Percutaneous tracheostomy tube equipment and trolley is prepared by the Senior Medical Officer.
- Use personal protective equipment (PPE) as per standard precautions.

6.3 Pre-Procedure

- Prepare bed areas ensuring the following are present:
 - Oxygen and suction present and in working order.
 - Tracheal dilators.
 - Selection of tracheostomy tubes (same outer diameter size as the ETT in situ).
 - **Sizing is manufacturer specific.** Shiley™ and Portex® **are not the same** so measure outer diameter.
- Inform the patient and / or significant others of planned procedure.
- If elective ensure the patient is fasted for six hours from food and two hours from water.
- Ensure the consent form has been signed.
- Prepare prescribed sedation and paralysing agents.
- Ensure the patient is in the supine position.
- Expose the cricothyroid region - clip hair as necessary.

6.4 Role of the Nurse

- Monitor haemodynamic and respiratory status during and post the procedure.
- Inform medical staff if any deterioration in condition.
- Liaise with medical staff regarding chest X-ray.
 - A post-procedure X-ray may assist in the identification of pneumothorax, pneumo-mediastinum or subcutaneous emphysema.
- If a newly inserted (<72 hours) PDT tube is dislodged, orotracheal intubation is the safest method to re-establish a patent airway.
 - No attempts must be made to re-advance the dislodged tracheostomy tube through the newly formed PDT tract,
- Monitor arterial blood gas (ABG). Within 1 hour of tracheostomy insertion.
- Document findings and interventions on unit specific chart / patient integrated notes.

6.5 Routine Tracheostomy Care

- A tracheostomy will require humidification treatment but eventually the body may adjust to the change and different forms of humidification may be utilised.
- The tracheostomy wound must be kept clean and dry to prevent post incision wound infection.
- Prior to assuming the care of the patient with a tracheostomy the nurse must be aware of:
 - Tracheostomy tube insertion date.
 - Reason for insertion.

- Type and method of insertion.
- Size and method secured.

7. Tracheostomy stoma site

- Monitor secretions (amount, colour and consistency).
- Humidification and suction requirements.
- Relevant clinical history including baseline vital signs and clinical findings.
- Oxygenation and oxygen administered.

Ensure the following are assessed at commencement of shift and as required:

- Patient comfort and respiratory status / airway patency.
- Check the tapes are secure.
- Cuff pressure (if cuff present).
- Assess the tracheostomy site and dressing for ooze, bleeding, secretions and attend to dressing as indicated.
- Inner cannula patent (as per inner cannula section).
- Bedside equipment additional to the above is present and in working order (i.e. suction catheters, sterile water within date).

8. Oral Care

- Oral care is important as it has a role to play in preventing healthcare associated infections.
- Dental plaque and the oropharynx can become colonized by bacteria and a 'biofilm' can develop on the inside of airway devices.
 - Secretions can also pool in the subglottic region.
- Normal oral airflow is disrupted when gas is directed through the tracheostomy and this leads to reduced evaporation of oral secretions, which subsequently accumulate in the mouth.
 - Patients who are able to should be encouraged to maintain their own oral hygiene by using a toothbrush followed by mouthwash.
- Incapacitated patients should have a daily assessment of their buccal mucous membranes to observe for bacterial, viral or fungal infections, skin tears or ulceration.
 - Aspirated infective saliva can contribute to respiratory problems.
 - If the patient has a dry mouth, consider artificial saliva.
- Any obvious dental problems should be assessed promptly by medical staff.
 - There is an increasing amount of data in the literature which suggests that simple measures such as teeth cleaning and intermittent removal of oral secretions can have a significant impact on hospital-acquired infections such as ventilator-associated pneumonia.

8.1 Specific oral care measures:

- Encourage self-care when possible.
- Patient's teeth/ dentures should be brushed with toothbrush and toothpaste at least twice a day.
- Utilise mouth wash twice per day.

- Patients may wear dentures with tracheostomies.
- Showering is permitted as clinically appropriate (with shower shield in situ).

9. Communication

- Assess and recognise the communication needs of the tracheostomised patient.
 - The impact of the loss of normal voice following tracheostomy should not be underestimated and whenever possible patients and their family should be prepared for this.
- The assessment starts preoperatively for elective patients.
- Multi-disciplinary assessment includes evaluation of the patient's ability to see, hear, touch, write, use and understand facial expressions such as smiling and blinking.
 - Assessment allows for problems to be quickly identified and immediate action can be taken to minimise consequences.
 - Following assessment, the nurse in liaison with the multi-disciplinary team will facilitate, design, implement and evaluate a communication care plan specific to the individual's needs and review as clinically indicated.
- Patient communication can be greatly enhanced following tracheostomy insertion by use of specialised products.
- The plan may include the use of mouthing words, coded blinks, closely watched facial expressions and gestures.
- Use of alphabet, picture and phrase books, electronic larynx and various types of tracheostomy tube.
- Document plan of communication in the patient integrated notes and Patient Care Plan.
- The multi-disciplinary team will identify strategies to facilitate speech. These may include:
 - A fenestrated tube.
 - Uncuffed tube.
 - Capping an uncuffed tube.
 - Decuffing a cuffed tube.
 - One way valves.
 - The valves will close off the airway whilst vocalising.
 - Inability to tolerate occlusion will make vocalisation impossible.
 - There needs to be some airspace left within the trachea.
- Patient must be medically stable.
- Secretions can be easily removed by coughing.
 - If secretions are too tenacious the valves may block, and the airway could become occluded.
 - Ability to vocalise with the current tracheostomy occluded with a finger.
 - Ability to have the cuff fully deflated without aspiration.
- Specialised speaking valves should be considered, as they can be beneficial in the decuffing / decannulation process for ventilated patients and decreasing secretions.

- Speaking valves should ideally only be used with an un-cuffed and fenestrated tube and only when the fenestrated inner cannula is in place.
 - It is possible to use with a non- fenestrated or even cuffed tube, providing that the cuff is deflated, and the patient has a fully patent airway above the tracheostomy.
- Unless a fenestrated tube is used, the size should not exceed 2/3 of the tracheal lumen.
 - Speaking valves are not to be worn whilst sleeping.
 - The nurse / Speech Pathologist is to teach communication techniques to help those with an uncuffed or fenestrated tube in situ.

9.1 Ways to promote communication for nurses/ward staff and family/carers

- Ask concise questions that require a “Yes” or “No” reply.
- Use simple hand signals or other appropriate patient agreed codes.
- Non-verbal communication can prove frustrating for patients.
- To avoid confusion, do not ask two questions at the same time.
- Be aware of all non-verbal communication signs to anticipate and react to problems and needs. For example, tapping the bed, waving, raising eyebrows with alerted expression.
- Provide patient with writing materials, white boards or magic slate.
- Equipment may be available from Speech Pathology or Occupational Therapy Departments.
- Develop lip reading skills.

Promotes a calm, unhurried approach when communicating with patients.

- Avoid finishing off sentences, repeat what has been understood then ask the patient to continue.
- Unless the patient has a documented reduction in hearing, do not raise voice when speaking.
- Reassure and discuss as appropriate.

10. Tracheostomy care bundle

In addition to the general care listed above, AKG utilises a Tracheostomy Care Bundle approach for the core management of Tracheostomies.

The bundle has six elements:

- [Safety Equipment](#)
- [Humidification](#)
- [Suctioning](#)
- [Tube Patency / Inner Tube Care](#)
- [Cuff Status/Pressure](#)
- [Tracheostomy Dressing](#)

10.1 Safety equipment

- Ensure the following equipment is present and in working order for routine and emergency use. Checked each shift by the Bedside nurse.

- Preferably set up on a dedicated trolley for easy access. **Note:** products required for the management of tracheostomies may be sourced from the ICU Physio and from ICU Shift Coordinator after hours and in emergencies.
 - Oxygen and suction equipment connected and in working order.
 - Suction catheters.
 - Tracheal dilator 1 (Level GPP).
 - Spare tracheostomy tubes (both the same brand - one the same size inner diameter and same length, and the other one size smaller but the same length).
 - If fenestrated tube is in situ, a non- fenestrated inner cannula must also be present to facilitate manual ventilation.
- Catheter mount (Cobbs connector).
- Manometer.
- Mapleson C circuit / Ambu bag
- Spare inner cannula (as per specific tracheostomy tube).
- Inner cannula cleaning equipment (saline and gauze or Uniperc cleaning swab).
- Velcro tapes or cotton tapes.
- Sterile 1L water (for suctioning).
 - Label with date and time of opening.
 - Discard if date not documented or if bottle opened greater than 24 hours.
- Disposable cup (for decanting sterile water for suctioning).
- PPE as per standard precautions.
- 10mL Luer lock syringe.
- This Patient has a Tracheostomy' poster at the patient's bedside, refer to [Appendix 3](#).

10.2 Humidification

Ensure adequate humidification. Check and document 2-4 hourly.

- Artificial humidification is to be utilised when a tracheostomy tube is in situ, for people requiring oxygen therapy – 'dry' oxygen must never be given to someone with a tracheostomy or laryngectomy.
 - The type of humidification will be determined by the needs of the patient.
 - During normal breathing, inspired air is warmed, filtered and moistened by ciliated epithelial cells in the nose and upper airways.
 - However, these humidifying functions are bypassed by a tracheostomy tube or laryngectomy and air inspired will be cold and dry.

Inadequate humidification can result in a number of physiological changes which can be serious to the patient and potentially fatal, including:

- Retention of viscous, tenacious secretions.
- Impaired mucociliary transport.
- Inflammatory changes and necrosis of epithelium.
- Impaired cilia activity.

- Destruction of cellular surface of airway causing inflammation, ulceration and bleeding.
- Reduction in lung function (e.g., atelectasis / pneumonia).
- Increased risk of bacterial infiltration.

10.2.1 Patient assessment to include:

- Frequency of suctioning and / or cleaning of inner cannula.
- Type and amount of secretions.
- Evidence of airflow via tracheostomy.
- Respiration rate and breathing sounds.
- Use of accessory muscles.
- Patient coughing (ineffective or excessive).
- Requirement for supplementary oxygen.
- Hydration status (e.g., diuretic therapy).

Adequate systemic hydration via oral, enteral or intravenous is required to ensure secretions remain easy to suction or expectorate

10.2.2 Skin integrity around tracheostomy

- All patients with a tracheostomy that are ventilated and on CPAP or BiPAP will receive warmed, humidified oxygen and / or medical air.
- Warmth and humidification settings to be monitored and documented in the patient integrated notes.
- Set up as per operator's manual.
 - Humidifier machine (e.g., Airvo or as appropriate to your area).
 - Change the humidification water and tubing every seven days.
 - Label the tubing with change date sticker.
 - Monitor the water level and change sterile water bags when empty and prn.
 - Avoid condensation in the humidification tubing circuit as it is a medium for bacterial growth especially pseudomonas growth.
 - Monitor the temperature of inspired gases.
 - Humidifier temperature to be set at 37 degrees to deliver a temperature ranging from 36.5 – 37.5 degrees Celsius at the tracheostomy site.
 - Liaise with senior nursing staff if temperature adjustment is required for patient comfort.
 - Contact the Ward Nurse Unit Manager (NUM) / Associate Nurse Unit Manager (ANUM) for replacement and management of equipment.
- Non-ventilated patients may receive humidification and warmth by tracheostomy nebuliser.

10.2.3 Heat and moisture exchanger ('Swedish nose')

- A Heat and Moisture Exchanger (HME) must be in use for patients with tracheostomy tube in situ particularly those with an inner cannula.
 - The amount of humidification will vary according to the patient's individual needs.

- Only to be used in patients with stable respiratory function ($\text{FiO}_2 < 40\%$) and low volume of secretions.
- Monitor the patient for signs of adequate humidification:
 - Check the HME every 1-2 hours (in situ, intact and not visibly soiled).
 - Change the HME daily (24 hours) or whenever soiled, as documented on the Patient Care Plan.

10.2.4 Administering a nebuliser to a non-ventilated patient through tracheostomy

10.2.5 Equipment

- Nebuliser mask with elbow connector.
- Nebuliser bowl.
- Tubing.
- Ensure oxygen source available.
- Sodium chloride 0.9% ampoule (sterile).
- PPE as per standard precautions.



Figure 4A: Nebuliser attached to tracheostomy.



Figure 4B: Tracheostomy nebuliser components.

10.2.6 Procedure

- Perform [hand hygiene](#).
- Inform patient of procedure and obtain consent as able.
- Ensure patient privacy as appropriate.
- Don PPE, perform [hand hygiene](#) and don gloves.
- Place 5mL sodium chloride 0.9% into bowl of nebuliser.
- Attach bowl to mask and tubing.
- Connect to oxygen supply.
- Remove garments or attachments from tracheostomy tube.

- Secure the mask around the patient's neck and attach to the air or oxygen source at a setting of 8-10 L/minute until nebuliser bowl is empty.
- Nursing staff to be able to visualise the patient throughout the administration of nebuliser.
- Storage and management of nebuliser device as per manufacturer's instructions
- Document on medication chart when nebuliser has been administered.

10.3 Suctioning

Check for secretions at least two hourly (Auscultate) and perform suction as required.

10.3.1 General information

- When suctioning the tracheostomy, it must be with a Tracheostomy length catheter.
- Assess patient to establish if suctioning is clinically indicated and observe for:
 - Audible or visible secretions.
 - Noisy and / or moist respirations.
 - Increased respiratory effort.
 - Prolonged expiratory breath sounds.
 - Restlessness.
 - Reduced oxygen saturation levels.
 - Increased and / or ineffective coughing.
 - Increased use of intercostal muscles.

10.3.2 Patient request

- In some clinical areas, measured depth suction may be performed.
 - This requires the operator to calculate suction depth according to tracheostomy tube manufacturer's instructions e.g., Kimberly-Clark products.
 - Patients may vocalize or signal a need to be suctioned.
- The frequency of suctioning is based on individual patient assessment and clinical need to avoid adverse effects.
- Consider factors such as:
 - Viscosity and quantity of mucus.
 - Neurological and muscular performance.
 - Presence of active cough reflexes and effort.
- Adverse effects of suctioning include tracheal trauma, suctioning induced hypoxia, infection, hypertension, cardiac arrhythmias and raised intracranial pressure.

10.3.3 Equipment

- Suction catheter – appropriate size.
 - The size of the suction catheter should be as small as possible, yet large enough to facilitate secretion removal.
 - It is generally recommended that the suction catheter should occlude less than half of the internal lumen of the endotracheal tube.

- PPE (gloves, goggles, apron and mask) as per standard precautions.
- Wall suction / portable suction machine checked and in working order.
- Sterile water 500mL bottle (for rinsing suction tubing).
 - Label with date and time of opening.
 - Discard if date not documented or if bottle opened greater than 24 hours.
- Consider pulse oximeter as required.
- If fenestrated tube is in situ, insert a non-fenestrated inner cannula before suctioning.
- The number of catheter passes when suctioning via a tracheostomy should be restricted to three and should be no longer than 15 seconds each suction, refer to the [AKG Suction Management Guideline](#).

10.4 Tube patency / inner tube care

Remove and check inner tube for secretion build up and clean as appropriate, 4-6 hourly.

10.4.1 Inner cannula management

Refer to [Types of Tracheostomy Tube section](#). The inner cannula (if present) must be:

- Checked for patency 2-4 hourly and for secretion build up.
- Cleaned and replaced a minimum of 4 hourly. (Can be done more regularly for patient with large volumes of secretions).
- Viscosity and volume of secretions precipitate more frequent checks and cleaning as indicated.
- **Note:** Determine if inner cannula is disposable.
- Retain the spare inner cannula from the insertion kit.
- Each patient must have a spare inner cannula of appropriate size (diameter) and length for use in emergencies or for cleaning purposes at the patient bedspace.
- The options of cleaning solution must be consistent with manufacturer's guide.
- The emergency equipment must be at the patient's bedside at all times.
- Monitor oxygen saturations before, during and after procedure
- Document care on the appropriate chart / Inner Cannula Care form.

10.4.2 Equipment

- Sterile dressing pack.
- Sterile saline or sterile water sachet.
- PPE (gloves, apron, goggles, mask) as per standard precautions.
- Spare inner cannula of appropriate size and length.
- **Note:** products required for the management of tracheostomies may be sourced from the ICU Shift Coordinator after hours and in emergencies.

10.4.3 Procedure

- Inform the patient of the procedure and obtain consent as able.
- Promote patient privacy.

- Open sterile dressing pack.
- Don PPE.
- Remove inner cannula from patient and place on sterile field.
- Insert clean / new temporary inner cannula into the outer tube.
- Add cleaning solution e.g., sterile water / sterile saline into dressing tray (as per manufacturer of tracheostomy).
- Refer to either “[regular length](#) or [extended length](#) inner tracheostomy tubes” for specific cleaning instructions for each type of inner cannula.
- Inspect the cleaned inner cannula to ensure it is patent and free from contamination before drying it for reuse in the same patient only.
- Repeat the procedure as necessary until the tube is clean.
- Dry the tube using Smiths medical Portex cleaning swabs.
- Appropriately dispose of equipment, waste and PPE.

Portex® tubes and UniPerc® tubes with inner cannulae are **not** to be cleaned with hydrogen peroxide.

10.4.4 Regular length inner tracheostomy tubes

- Submerge the inner cannula into sterile water or sterile saline - thickened secretions can be loosened by using the adjustable flange inner cannula cleaning swab.

For Shiley™ tracheostomy tubes remove the red temporary inner cannula and reinsert cleaned white inner cannula. Refer to the [Inner Cannula Tracheostomy Tube Management Table](#)

- For heavily soiled Shiley™ inner cannulas, consider cleaning with half strength hydrogen peroxide 0.5%.
- Following reinsertion of white inner cannula, clean red temporary inner cannula using the same procedure.
- The same dressing pack can be utilised by placing sterile water / sterile saline into clean division of dressing tray.
- Dry and store as per Storage of spare Inner Cannula section.
- If heavily soiled inner cannula that cannot be cleaned, discard and replace with identical inner cannula.

10.4.5 Extended length inner tracheostomy tubes

- Use only the specific cleaning swab provided for the cleaning of UniPerc adjustable flange tracheostomy inner cannula – refer to [Figure 5B](#) below.
- Remove the “dirty” cannula by means of the blue ring pull on the inner cannula and replace with a new / cleaned cannula **at least eight-hourly**.
 - A slight snap should be felt indicating proper positioning on insertion of clean inner cannula.
 - This inner cannula will remain in situ until the next change.
- Do not soak inner cannula.

- Clean the inner cannula with sterile water/sterile saline by inserting the moist UniPerc cleaning swab and remove it with a twisting action to clean the inner lumen.
- Hold the clean inner cannula up to the light.
 - If it has become translucent or lighter in sections then discard and replace. (Durability of the inner cannula may deteriorate after 7-10 days of continuous use).
- Dry and store as per Storage of Spare Inner Cannula section.

Note: Blue ring on inner cannula (below)



Figure 5A: Portex UniPerc adjustable flange tracheostomy inner cannula with blue ring pull



Figure 5B: Inner cannula UniPerc cleaning swabs

10.4.6 Storage of Spare Inner Cannula

- **Note:** Inner cannulas are **not** to be stored wrapped in gauze.

10.4.7 Regular length inner tracheostomy tubes

- Following appropriate cleaning of inner cannula, place it in a clean, labelled disposable storage container with lid (e.g. denture container).
- Labelled storage container is to be replaced a minimum of every 24 hours and if visibly soiled.

10.4.8 Extended length inner tracheostomy tubes

- Following appropriate cleaning of inner cannula, place it in a cleaned (detergent wiped) and labelled kidney dish.

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- Place this into a biohazard bag – leave unsealed until the inner cannula is fully dried, then seal the bag.
- The bag and kidney dish are to be replaced a minimum of every 24 hours and if visibly soiled. Date and time the container, to monitor compliance.

Table 1: Inner cannula tracheostomy tube management table

Tracheostomy / Type	Inner Cannula Management	Inner Cannula
 Shiley™ (Regular length)	Maximum recommended period of use is 30 days, Replace if damaged. Note: Red temporary inner cannula is only for use for periods less than 10 minutes while cleaning white inner cannula. Refer to: Regular Length Inner Cannula Management Procedure and Storage section. Remove inner cannula and replace with clean inner cannula at least every eight hours.	 Red temporary inner cannula  White inner cannula Note: All inner cannulae with twist lock connectors should not be used on any other tube because they are fitted to the exact length of individual tubes.
Portex Blueline® (Regular Length)	Maximum recommended period of use is 30 days. Replace if damaged. Refer to: Regular Length Inner Cannula Management Procedure and Storage section. Remove inner cannula and replace with clean inner cannula at least every eight hours.	Pt must have two at bedside.
 Portex UniPerc® (Extended length)	Maximum recommended period of use is 29 days. Replace if damaged. Refer to: Extended length Inner cannula management / procedure and storage section.	

Tracheostomy / Type	Inner Cannula Management	Inner Cannula
Note: Always refer to manufacturers' instructions that come with the tracheostomy tube if unsure of cleaning instructions.		

10.5 Cuff status / pressure

10.5.1 Cuff management

- A cuffed tracheostomy tube may be required to facilitate ventilation and for those patients with impaired swallowing.
- Cuff patency assessment includes the use of:
 - Minimum obstructive volume (MOV) or
 - Minimum leak technique (MLT).

Figure 6: Cuff manometer



Cuff pressures vary but **must not** be greater than 40cm H₂O

- The goal of cuff inflation is to achieve a seal between the cuff and the trachea with a pressure great enough to prevent aspiration, but not so high as to cause tracheal damage (e.g. tracheal stenosis).
- Utilise a cuff manometer to measure cuff pressure:
 - At the beginning of each shift or every 8 hours to determine a base line measurement.
 - After manipulation of the tracheostomy tube or significant patient movement .
 - Repeat measure as clinically indicated, i.e. if an air leak.
- Cuff pressure should be maintained between 20-30cm H₂O
- If an air leak persists despite reinflation of air to a maximum of 25cm H₂O contact the Medical Officer immediately and prepare equipment for a tube change.
- An air leak may be suspected if audible air leak around the tube (if ventilated) or patient has persistent coughing.
 - An air leak is identified by a persistent decrease in the cuff pressure.
- Routine cuff deflation is not recommended unless the patient is being prepared for decannulation.

- Routine cuff deflation to relieve pressure is associated with increases in aspiration and hypoxaemia and does not significantly affect wall pressure.

10.5.2 Process

- Attach manometers side port to air inlet valve of tracheostomy.
- Read the gauge.
- Document reading in the unit specific chart.

10.6 Tracheostomy dressing

Check and change dressing at least every 24 hours or if it is dirty.

- Note: 2 staff should be present when changing the tracheostomy tape/dressing

10.6.1 Care of stoma

- Assess tracheostomy stoma (edges of stoma) a minimum of daily and as clinically indicated.
 - Tracheostomy stomas are a potential avenue for respiratory tract infection and potential site for peristomal infection.
 - The tracheostomy wound should be kept clean and dry to prevent post incision wound infection.
 - Change dressings more frequently if they become wet, soiled or there are increased secretions to decrease the risk of infection and impaired skin integrity.
 - In acute care settings, saline and gauze are considered optimal for stoma management.
 - Assess for signs of infection or irritation which include:
 - Bleeding.
 - Infection.
 - Pressure from tracheostomy flanges.
 - Secretions / crusting.
 - Redness.
 - Drainage.
 - Crusting.
 - Odour.
 - Pain.
 - Irritation.

10.6.2 Dressing change – equipment

- Trolley.
- Standard dressing pack.
 - Do not use cotton wool to cleanse around stoma.
 - Note: a standard dressing pack contains three sterile forceps but up to four sterile forceps will be required for the procedure.
- Extra sterile forceps as required.
- Extra gauze swabs as required.

- 1 pair sterile surgical scissors.
- Sodium chloride 0.9% (or prescribed solution).
- Rubbish bag.
- Tracheostomy dressing - dressing options include gauze (in tracheostomy dressing pack) and fenestrated foam products.
- Two pieces white tape 40-45cm long or Velcro ties (Note: Change and discard if tape or Velcro becomes soiled / wet).
- PPE gloves as per standard precautions.

10.6.3 Procedure

- Open the dressing pack and additional sterile equipment maintaining asepsis.
- Apply PPE.
- Perform [hand hygiene](#) prior to donning gloves.
- Moisten gauze swabs with sterile sodium chloride 0.9% - wring to remove excess solution (should be wet but not dripping).
 - **Do not** cut gauze as fibres fray and may become inhaled or trapped under the tracheostomy flange.
- Use sterile forceps to remove dressing and discard (if one in use).
- Place sterile dressing towel on patient's chest.
- Using sterile forceps swab flange of tube ensuring all debris removed.
 - Avoid the introduction of fluid into the tracheostomy which increases the risk of infection.
- Dry in the same manner.
- Swab tracheostomy stoma whilst using forceps to support / lift the flange as this enables stoma to be visualised for assessment.
- Dry in same manner.
- The skin must be kept dry and liquid creams avoided.
- Repeat as necessary to clean tracheostomy
- Apply dressing (may not be required if long term tracheostomy) by supporting the tracheostomy flange with a forceps, pull the dressing through one side then bring the other side through.
- **Note:** movement of the tube can cause tracheal irritation.

10.6.4 Securing the tracheostomy

The tracheostomy tube is to be secure at all times.

- This is to be done by the use of Velcro straps, cotton tapes or sutures.
- Securing and positioning of the tracheostomy tube must prevent dislodgement and maintain alignment particularly during patient repositioning / suctioning.
- Cotton tapes may be used if patient is agitated, as Velcro strap may stretch if pulled.
- Velcro straps / cotton tapes to be replaced during tracheostomy dressing or more frequently if soiled or wet.
 - Single patient use device.
- If Velcro straps / cotton tapes require changing, request assistance to secure the tracheostomy tube, or

- If one person technique, leave existing Velcro straps/cotton tapes in situ until new Velcro straps / cotton tapes are secured then remove existing tapes.
- Tapes x 2 to be applied.
- Put fold through hole in flange, one each side.
 - Tapes need to be folded at the 1/3rd point with 2/3rd remaining long enough to reach around the patient's neck.
- Take both threads through the loop to form a slip knot.
 - No bows.
- Take the long thread behind the neck and tie it to the short thread on the other side, using a double reef knot.
- It is not recommended that tapes / straps be tied over a muscle flap, as this may cause constriction of blood flow.
- Repeat the procedure for the opposite thread.
- Cut old threads and remove the loop from the hole in the flange.
 - Ensure the cuff port is avoided.
- Assess the tightness of the tapes / straps.
 - A maximum of two fingers should slip between the tape / strap and the patient and feel firm.
- Appropriately dispose of waste and PPE.

Figure 7: Application of Velcro straps



10.6.5 Potential complications

Complications can be divided into those associated with:

- Insertion of the tracheostomy (surgical or percutaneous) such as pneumothorax, surgical emphysema and loss of the upper airway.
- Those which arise following the procedure (usually blocked or displaced tracheostomy tubes).
- Later complications such as:
 - Blocked / obstructed tube.
 - Displaced /dislodged tube.
 - Haemorrhage.
- Tracheal damage / ischaemia
 - Pneumonia.
 - Local infection.
 - Cuff leak.

10.6.6 Blocked / obstructed tube

- Blocked tubes may occur at any time, especially if secretions become thick, the secretions are not managed appropriately (suction) and humidification is not used.
- The upper airway is designed to moisten and warm inhaled air before it reaches the lungs.
 - Cilia are small hair like protrusions that line the respiratory tract; the function of the cilia is to prevent infection within the respiratory tract by moving mucus and other particles away from the lungs.
- Inserting a tracheostomy tube bypasses these natural mechanisms, which mean the lungs will receive cool, dry air.
 - Dry air entering the lungs may reduce the motility of the secretions within the lungs and may reduce the function of the cilia.
- In addition, the patient may not be able to cough and/or clear the secretions from their airways through the tracheostomy.
 - This may cause the tracheostomy to become blocked by these thick or dry secretions.
 - Blocked tracheostomy tubes can be minimised by careful humidification, tracheal suction and inner tube care.
- Herniation of cuff over end of tube and tracheostomy tube sitting against tracheal wall may also obstruct a tracheostomy.

10.6.6.1 Management

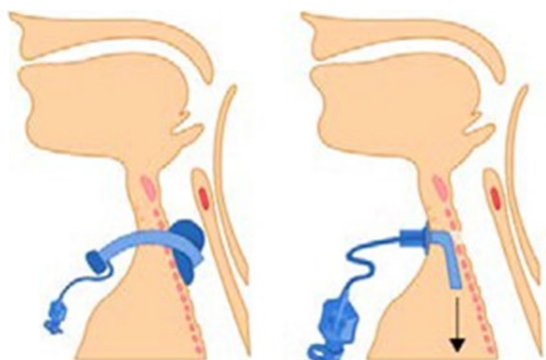
- Suction the tracheostomy tube to remove obstruction.
 - If suctioning does not remove obstruction then activate medical emergency.
 - Refer to [Appendix 1: Emergency Tracheostomy Management – Patent Upper Airway Flowchart](#).

Ensure emergency equipment is always at hand, as a blocked tube may lead to respiratory arrest.

10.6.7 Displaced / dislodged tracheostomy tube

- The tracheostomy tube can become partially or completely displaced.
 - The tube may migrate out of the stoma or into the soft tissue of the neck.
 - The tube may become displaced by coughing, because of its weight or the weight of attached breathing circuits, or by patient interference.
- Partial tube displacement is more dangerous as it is not always visibly obvious that there is a problem with the tube.
- In order to keep tracheostomy tubes in position they must be secured carefully and monitored.
- Any concerns raised by the patient or nursing staff must be promptly investigated.

Figure 8: Tracheostomy tube dislodgement



10.6.8 Management

- Seek assistance from senior staff and call a medical emergency team (MET).
- If a newly inserted (<72 hours) PDT tube is dislodged, orotracheal intubation is the safest method to re-establish a patent airway.
- No attempts must be made to re-advance the dislodged tracheostomy tube through the newly formed PDT tract.
- Activate a medical emergency as clinically indicated.

10.6.9 Haemorrhage

- It is common for some bleeding to occur after a tracheostomy has been performed.
 - This usually settles with a few days; however, these points must be considered:
 - Bleeding can occasionally be significant or even catastrophic.
 - Bleeding can be from the trachea, stoma or surrounding tissues and can be due to direct trauma of the tissues, puncture or injury to adjacent blood vessels or the tube or cuff eroding into surrounding tissues or vessels over time.
 - Bleeding can also come from the lungs themselves and become evident through tracheal suction.
 - Bleeding problems are compounded in a patient with a coagulopathy.
- A trachea-innominate fistula can occur if the tube erodes into the innominate artery and may present as a pulsating tracheostomy.
 - This is a rare complication but is associated with lower placement of the tube in the trachea.
- The hallmark is a warning or 'sentinel' bleed.
- Any haemorrhage should prompt a fibre optic inspection of the trachea.
- If an arterial bleed is suspected, this must occur immediately with an experienced surgeon and resuscitation measures available.
- Arterial haemorrhage can become rapidly fatal.
- Hyperinflation of the tracheostomy tube cuff or an endotracheal tube cuff may help to tamponade the bleeding point, prior to definitive surgical management.

10.6.10 Management

- Report any excessive or new pulsation to the Medical Officer immediately.
- Check that cuff pressure is within normal limits.

- Over-inflated cuff can exert pressure on the innominate artery.
- Bleeding from the tracheostomy must be treated as a medical emergency.
- Rupture / erosion of the innominate artery has a high risk of mortality and will require surgical repair.
- Initiate emergency response.
- Any haemorrhage must prompt a fibre optic inspection of the trachea.
- Suction tracheostomy to remove excess blood, secretions and clots.
 - Secretions of any kind can lead to airway obstruction as a result of blood clots and secretions.

10.6.11 Tracheal damage / Ischaemia

- Damage to the trachea may be caused by the pressure of the inflated cuff pressing on the mucosa of the trachea.
 - The capillary pressure in the tracheal mucosa is around 20cm H₂O and consistent cuff pressure above this limit will risk ischaemic damage to the trachea.
 - This situation may be made worse by critical illness and hypotension, which will reduce the capillary perfusion pressure.
- Direct mucosal damage can also occur by poor tracheal suctioning techniques, ill-fitting tubes, or excessive movement of the tube within the trachea.
- Modern tracheostomy tubes have low-pressure cuffs which extend over a greater surface area, however over inflation must still be avoided.

10.6.11.1 Management

- The pressure in the cuff should be just adequate to prevent air leakage and seal the airway against aspiration (maintain cuff pressure between 20 and 30cm H₂O).

10.6.12 Pneumonia

- A build-up of secretions may lead to consolidation and even collapse of some areas of the lung, and thus contribute to pneumonia.
 - The presence of a foreign body in the airway will hamper normal physiological defence mechanisms and particulate matter, oral secretions, gastric contents and bacteria can be aspirated past any cuff into the airway.
- These risks can be minimised by careful humidification, tracheal suction and inner tube care, and may be helped by suctioning above the cuff with specific subglottic suction tubes.
- Aspiration of gastric contents may also lead to pneumonia - this can occur with patients who are unable to swallow safely.
- Any patient who you suspect may have aspirated will need to have a formal swallowing assessment.

10.6.13 Management

- Determine cause of aspiration and rectify if possible e.g. increase tracheostomy tube size, altering feed regime, ensure bed elevation during feeds.
- Maintain head of bed elevation to at least 45° when feeding patient and for 30 minutes post-feed.

10.6.14 Local infection

- The stoma is an open surgical wound, and like any wound is at risk of infection.

10.6.15 Management

- A stoma should be treated as a surgical wound and cared for appropriately.
- Vigilant observation, keeping the wound clean and dry with regular dressing changes will help to reduce the risk of infection.
- Prompt swabbing / specimen collection and appropriate treatment will assist to identify and minimise the impact of local infection.
- As the stoma is a wound opening directly into the respiratory tract there is potential for the lower respiratory tract to become infected.
- Poor suction technique may also increase the incidence of infection.

10.6.16 Cuff leak

Identified by:

- Reduction in cuff pressure when tested.
- Patients may require an inflated cuff to facilitate ventilation or to protect the airways from aspiration.
- Increasing patient distress and coughing more than usual.
- Possible causes:
 - Faulty valve.
 - Hole in cuff.
 - Accidental cutting / pulling of the cuff port.

10.6.17 Management

- Initially re-inflate cuff and recheck within 30 minutes or as required.
 - If cuff is deflated in preparation for de-cannulation then an inflatable cuff may not be required.
- If not maintaining pressure liaise with the MO, consider urgent tracheostomy change.
 - Refer to the [Tracheostomy Tube Change section](#)
 - Assessment of the patient's need for an inflated cuff to be undertaken.

10.6.18 Tracheostomy red flags

- Tracheostomy red flags may be clues that a problem has or is about to occur and needs to be acted upon (warning signs).
- Red flags can be divided into different categories:
 - Airway.
 - Breathing.
 - Specific tracheostomy flags.
 - General flags.

10.6.19 Airway red flags

- If the patient has a cuffed tracheostomy correctly sited in the trachea, no gas should escape through the mouth.

- If the patient is talking to you, or audible air leaks or bubbles of saliva are seen or heard at the mouth or nose, then gas is escaping past the cuff.
 - This may imply that the cuff is damaged, or the tube tip is not correctly sited.
- Grunting, snoring or stridor is also signs that there is an airway problem.

10.6.20 Breathing Red Flags

- Listening to the patient, or observation of the patient or instrumentation, may show that the patient:
 - Is not breathing (apnoea), which is detected by capnography or clinically.
 - Has difficulty in breathing (or with ventilation), which may be reported by the patient or observed clinically:
 - Accessory muscle use.
 - Increased respiratory rate.
 - Higher airway pressures.
 - Lower tidal volumes.
 - Has hypoxia.
 - Is making whistling noises or has noisy breathing.

10.6.21 Specific Tracheostomy Red Flags

- Careful observation may show that the patient:
 - Has a visibly displaced tracheostomy tube.
 - If this is an adjustable flange tube, check to see where it was last positioned.
- Has blood or blood-stained secretions around the tube - a recently performed or changed tracheostomy bleeds a little, but if in doubt, it must be assessed.
- Reports increased discomfort or pain.
- Requires a lot of air to keep the cuff inflated, which may be because:
 - The cuff is damaged or has an air leak (in which case, it needs to be replaced).
 - The tube may be displaced, and the cuff needs hyper-inflation to keep it 'sealed'.

10.6.22 General red flags

- Any physiological changes can be due to an airway problem.
- Specifically changes in:
 - Respiratory rate.
 - Heart rate.
 - Blood pressure.
 - Level of consciousness.
 - Anxiety, restlessness, agitation and confusion may also be due to an airway problem.

10.6.23 Tracheostomy tube change

First Tube Change

- It is the responsibility of the surgeon or team who performed the initial insertion procedure to conduct the **first** tracheostomy tube change; or the treating team to consult with the Surgeon / team who performed the initial insertion procedure regarding management prior to first tube change.
- The first tracheostomy change (surgical and percutaneous) is to be performed during business hours and preferably as early in the day as possible.
- If a difficult tube change is anticipated then a clinician experienced in upper airway management (including endotracheal intubation) and a clinician or practitioner proficient and experienced in managing tracheostomy tubes should be present.
 - All relevant equipment and emergency anaesthetic drugs should be available.
 - Plan for loss of airway and failure to cannulate the tracheostomy.
- If tracheostomy tube change is deemed complex by insertion procedure list, or the tracheostomy tube is unfamiliar to ward staff, seek assistance from staff experienced in specific tracheostomy tube changes. Uniperc tracheostomy tube changes are not to be performed by nursing staff.
- **For subsequent tracheostomy tube changes** that are deemed appropriate for nursing staff to perform: a minimum of two nurses must be present during tube change / decannulation one of whom must be experienced in the procedure.
- Medical staff to be present on the ward and informed that procedure is being carried out.
 - Ensure specified emergency equipment is present at the bedside.
- Products required for the management of tracheostomies may be sourced from the ICU Shift Coordinator after hours and in emergencies.

10.6.24 General points

- Double lumen tube (with inner cannula) is changed every 28 days.
- Manufacturers recommendation is that usage must not exceed 29 days.
- Two people are required to perform a tracheostomy tube change], one of whom must be experienced in the procedure.
- The first tube change to be carried out no sooner than 72 hours post insertion.
- Emergency equipment at bedside must be present and in working order (including tracheal dilators).
- For the first tracheostomy tube change:
 - A MO with advanced airway skills must be present for the procedure.
- Refer to specific clinical area regarding personnel to perform procedure.
- Changes of outer tracheostomy tube must be performed by a practitioner who is trained in the technique.
- Subsequent changes can be undertaken by nursing staff, one of whom must be experienced in the procedure
- Patient should be nil by mouth / fasted for six hours from solids, and two hours for water before an elective tube change, or if a nasogastric (NG) tube is present, this must be aspirated to empty the stomach.

Particular caution needs to be taken in the following circumstance:

- First Tube change.
- Difficult previous tubes change.
- Patient has a known obstructed or difficult upper airway (this is the 'back up' airway in case of difficulties with the stoma).
- Early changes (within 2-4 days of surgical tracheostomy and 7-10 days of percutaneous tracheostomy).
- Patients with a large or short neck.
- Obesity.
- Patients requiring high levels of ventilator support.
- Patients requiring high concentration of inspired oxygen.
- Patients with tumours surrounding the tracheostomy tract.
- Patients with significant granulation tissue around the stoma site.
- Agitated or combative patients.

10.6.25 Equipment

- Standard dressing / tracheostomy pack.
- Tracheostomy fenestrated dressing.
- Sterile gloves.
- Tracheostomy tube (correct size and one size smaller) smaller size in case difficulty inserting same size tube.
 - Store in packaging on bottom of trolley unless needed.
- Sterile syringe (if cuffed tube).
- Sterile scissors
- Sodium chloride 0.9% 30mL sachet.
- Sterile water based lubricant.
- Stitch cutter.
- Sterile McIndoe forceps.
- Sterile dissecting forceps.
- Sterile gauze swabs. Insufficient swabs in tracheostomy dressing pack to complete dressing.
- Pulse oximetry machine.
- PPE as per standard precautions.

Note: Products required for the management of tracheostomies may be sourced from the ICU shift coordinator after hours and in emergencies.

10.6.26 Preparation

- Patient must be nil by mouth / fasted for six hours from solids, and two hours for water before an elective tube change, or if a NG tube is present, this must be aspirated to empty the stomach to minimise risk of aspiration.
- Advise the MO of the tube change.
 - The MO to be readily available during procedure.

- Ensure oxygen and suction are in working order.
- Suction catheter / Yankauer connected to suction tubing.
 - Turn suction 'ON' and place at head of bed.

10.6.27 Medical staff undertaking change

- Prepare tracheostomy dressing pack and add other required equipment using aseptic technique.
- Don PPE and sterile gloves.
- Inflate / deflate cuff on new tube (if present) to evaluate for patency of cuff using sterile syringe provided in pack, to check for air leaks and spontaneous inflation / deflation.
- Attach sterile tapes to tube, fold 1/3 to 2/3 and loop through holes in flange.
 - Tube is to be secure at all times to prevent against accidental dislodgement of tube. **Note:** If using Velcro tapes this is not necessary.
- Lubricate the tube with lubricant product to facilitation insertion and maintain sterility.
- Prepare gauze swabs by moistening and then wringing to remove excess solution.
- If swabs too wet solution can access lungs via stoma.
- Place sterile dressing towel on patient's chest.
 - Place sterile forceps on dressing towel.
 - These forceps are used to hold the flange during cleaning of stoma.
- Check that introducer moves out of tube freely.
- Complete tracheostomy dressing.

10.6.28 Nurse procedure

- Don PPE.
- Apply pulse oximeter to patient.
- Suction the tracheostomy if required and oropharynx.
 - Remove excess secretions that may increase risk of aspiration.
- Pre-oxygenate as indicated.
- Aspirate nasogastric or gastrostomy tube (if in situ) to minimise the risk of aspiration
- Position patient supine or upright with neck slightly extended unless contraindicated.
 - This facilitates removal and insertion of tube.
- Perform suction simultaneously whilst deflating the cuff of the tube in situ, then remove Velcro straps / tapes.
- Place thumb and forefinger either side of the termination on flange to hold in place until ready to remove.
- Remove any attachments (oxygen, ventilators, humidivent).
 - Place on disposable towel
- When person performing tube change is ready they will communicate the timing of the tube change.
- Remove tube with a firm out and downwards movement as patient breathes out.
- Resistance may be felt as the cuffed portion of the tube exits the stoma.

- Appropriately dispose of equipment, waste and PPE.
- If unable to insert new tube, activate medical emergency.
- Secure new tube using tapes or if using Velcro hold tube in place whilst assistant attaches Velcro straps.
 - The tube is to be manually held in place during the tube change and until it can be adequately secured.
- Replace the dressing.
- Assess respiratory status of the patient, specifically respiratory rate and effort, skin colour, oxygen saturation and air entry.
 - It is imperative to assess both patency of the airway and associated effectiveness of respiration.
- Document tube change including details of adverse findings and interventions in patient integrated notes.
- Update the Patient Care Plan to reflect date of next scheduled tube change.
- Appropriately dispose of equipment, waste and PPE.
- Complete tracheostomy tube sticker in patient integrated notes.

11. Weaning and de-cannulation

11.1 General information

Decannulation is a multi-disciplinary decision planned by specialist team lead by the ICU Consultant.

- Patients who have recovered from a brief illness may tolerate immediate tracheostomy tube removal.
- Decannulation is to be performed by a doctor or nurse deemed experienced in the management of tracheostomies or under the direct supervision of an experienced doctor or nurse.
- The patient is ready for decannulation when the reason for tracheostomy tube insertion has resolved and they are able to maintain a patent airway.
- Two nurses to be present during the decannulation procedure.
- Weaning to be performed in collaboration with the MDT and lead by the ICU consultant.

11.2 Criteria for de-cannulation

- Adequate spontaneous breathing for 24- 48 hours (whilst capped and cuff deflated).
- Patent upper airway.
- Ability to maintain and protect airway spontaneously.
- Free from ventilator support with adequate respiratory function.
- Haemodynamically stable.
- Absence of fever or active infection.
- Strong consistent cough (able to cough into mouth).
- Control of saliva + / - a competent swallow.
- Clinical stability.

- Patient consistently alert prior to decannulation.

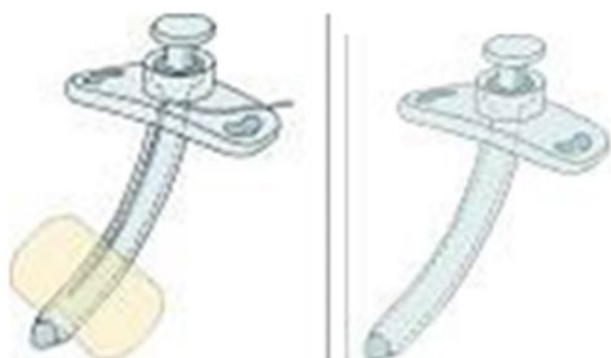
11.3 De-cuffing regimes

- Decuffing regimen is determined by the multidisciplinary team and led by the ICU consultant.
 - If the patient's ability to cope improves or deteriorates with their current regimen their decuffing regimen to be altered as clinically indicated.

11.4 Fast track:

- The cuff is to remain deflated for 24 hours.
- Meets criteria for decannulation.
- Immediate decuffing for 24 - 48 hours consecutively.
- Extubation after 24 - 48 hours if tolerated.

Figure 9: Cuffed and non-cuffed tubes



11.5 Decannulation procedure

- This is a two person procedure as per tracheostomy tube change.
- Decannulation is to be regarded procedurally as a tracheostomy tube change, due to the potential risk of failure and the need to reinsert a tracheostomy tube.
- The patient should be nil by mouth/fasted for six hours from solids, and water for two hours, or if a NG tube is present, this must be aspirated to empty the stomach before decannulation.
- Aspirate the nasogastric or gastrostomy tube (if in situ).
- Apply a dressing to stoma site immediately post-decannulation.

11.6 Post-decannulation

- Monitor oxygen saturations, pulse respiratory rate and signs of respiratory distress:
 - 15 minutely for two hours, and
 - then at least every ½ hour for two hours or as indicated post decannulation, then
 - a minimum of 4/24 for 48 hours.
- Oxygen saturation and heart rate are the most reliable indicators of the patient's tolerance and must be taken during the entire decuffing / decannulation phase.
- Monitor the patient closely for 48 hours after removing the tube to ensure tolerance from decannulation.

- Observe the dressing TDS.
- Dressing intact for 24 hours.
- If becomes loose – resecure.
- Remove the dressing at 24 hours and assess to determine management.
- Review the stoma every 24 hours:
 - Reduction in stoma size.
 - Wound healing.
 - Redness / swelling.
 - Irritation.
- Document findings in patient integrated notes and contact MO as clinically indicated.
- A smaller tracheostomy tube and tracheal dilators should be available at the bedside for 24-48 hours after removal.
- The need to replace a tracheostomy tube usually arises within 48 hours of decannulation e.g. respiratory distress.
 - If this occurs treat as an emergency situation.

11.7 Transporting a Patient with a Tracheostomy

- Only a nurse experienced at tracheostomy tube management may escort a patient with a tracheostomy when not accompanied by a MO.
- There is always a risk that a tracheostomy tube may become blocked, dislodged or removed completely.
- Inform the Shift Coordinator when departing the ward / unit.
- Receiving area is to be notified that the patient has a tracheostomy in situ.

11.8 Transfer Destination (Intra or Inter Hospital)

- Availability / access to equipment. E.g. request Twin-o-Vac suction and / or oxygen equipment.
 - Patients who cannot clear their own secretions and require frequent suctioning are at greater risk of respiratory complications.
- Patient clinical condition and acuity.
- Emergency airway equipment is to be transported with the patient.
 - Suction catheters.
 - Yankauer sucker.
 - Tracheal dilators (for trained medical staff use only).
 - Tracheostomy ties and dressing
 - Two tracheostomy tubes:
 - One same size, and
 - One smaller size.
 - Size appropriate to tube in situ plus smaller one.
- One full oxygen cylinder if oxygen therapy required by the patient with suction unit attached.
- PPE.

11.9 Patient Assessment and Preparation Prior to Transport

- Patency of tracheostomy - suction patient as required.
- Assess and record oxygen saturation, respiratory pattern, rate and work of breathing as clinically indicated.
- Baseline observations to compare alterations in condition.
- Note: patients with an established tracheostomy who are haemodynamically stable may not require observations prior to transfer.
- Ensure the tapes are secure.
- Prepare additional monitoring equipment as indicated by clinical condition.

12. Legislative context

The following documents are required to give effect to this procedure [i.e. the documents included are mandatory]:

- Health Department of Western Australia (HDWA) [Consent to Treatment Policy](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Infection Prevention and Management Policy EMHS: 34](#)
- EMHS [Consent Policy EMHS: 5](#)
- Armadale Kalamunda Group (AKG) [Hand Hygiene Policy](#)
- AKG [Recognition and Response Acute Deterioration Policy](#)

13. Supporting information

The following documents support the implementation of this procedure:

- [AKG Airway Management – Adult Procedure](#)
- [AKG Suction Management Guideline](#)

14. Compliance, monitoring and evaluation

Compliance with this policy will be evaluated by the Recognising and Responding to Clinical Deterioration Working Group's audit processes, and by routine clinical incident monitoring via the DATIX-CIMS system.

15. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.5: The health service organisation has processes to apply standard and transmission-based precautions that are consistent with the current edition of the Australian Guidelines for the Prevention and Control of Infection in Healthcare, and jurisdictional requirements
- Standard 6 – Action 6.4: The health service organisation has clinical communications processes to support effective communication when:
 - Identification and procedure matching should occur
 - All or part of a patient's care is transferred within the organisation, between multidisciplinary teams, between clinicians or between organisations; and on discharge
 - Critical information about a patient's care, including information on risks, emerges or changes

- Standard 8 – Action 8.10: The health service organisation has processes that support timely response by clinicians with the skills required to manage episodes of acute deterioration

16. Glossary

The following definitions are relevant to this policy.

Term	Definition
Tracheostomy	An artificial opening into the trachea through the neck
Glossectomy	Removal of the tongue
Cricothyroidotomy	Creation of an opening in the Cricothyroid membrane to facilitate gas exchange

17. Policy contact

Enquiries relating to this procedure may be directed to:

Title: Nurse Unit Manager
 Department: Intensive Care Unit
 Email: muralee.muthukrishnan@health.wa.gov.au

18. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

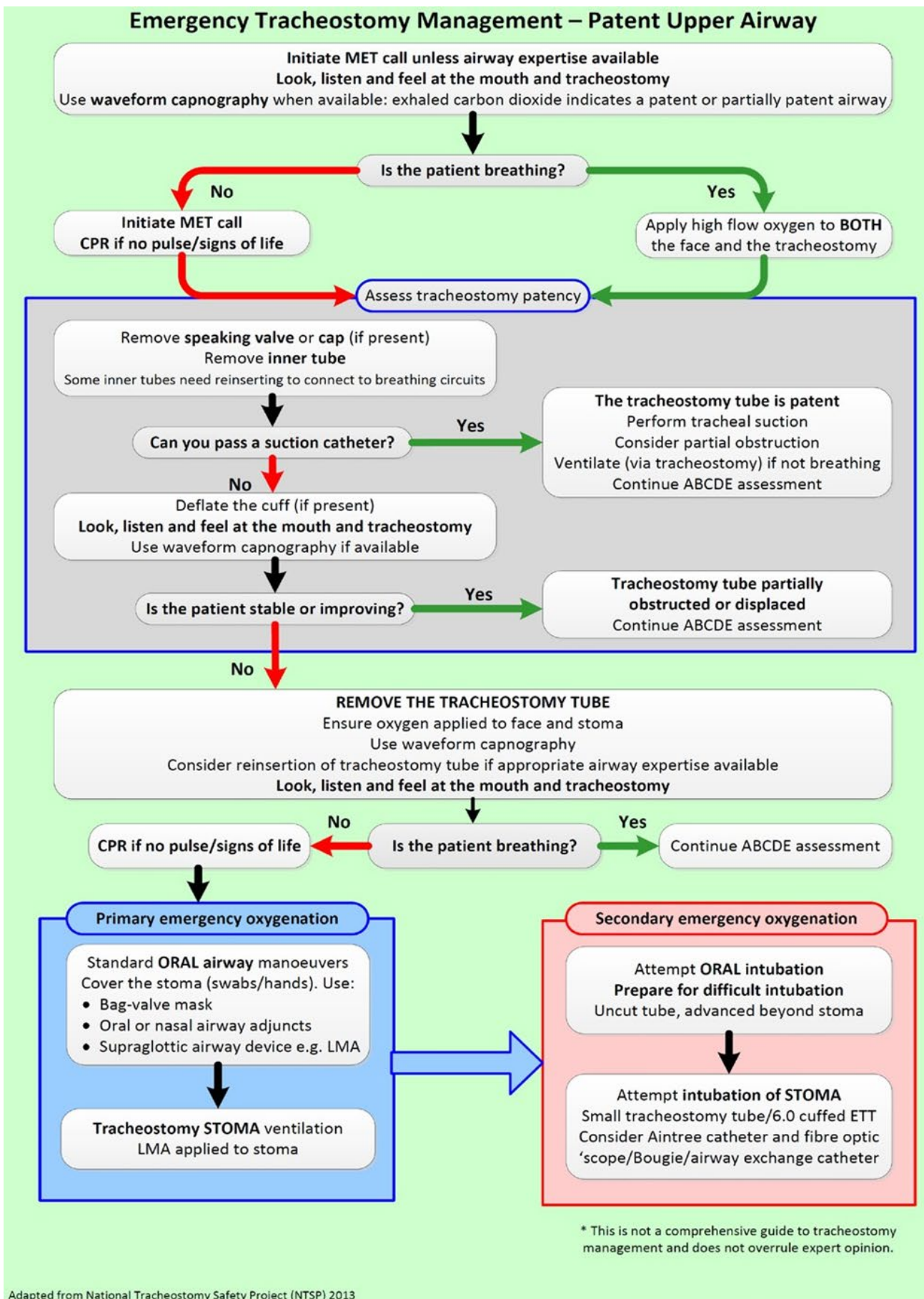
Version	Effective from	Effective to	Amendment(s)
V3	25/05/2023	25/05/2026	Scheduled review

19. Authorisation

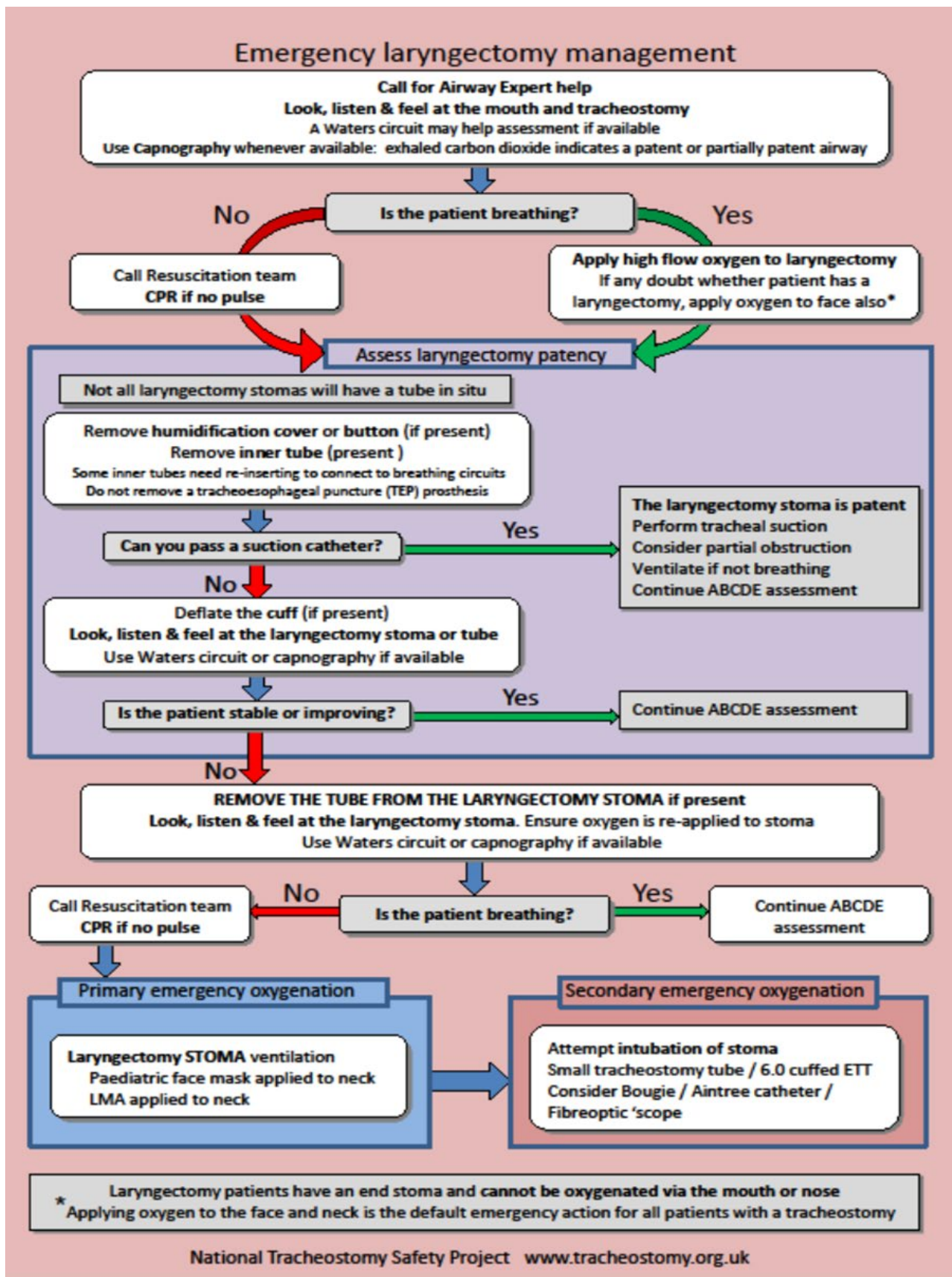
This procedure has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Xin Chau
Authorisation date	A/Director Clinical Services
Published date	25/05/2023

Appendix 1 Emergency Tracheostomy Management Flowchart



Appendix 2 Emergency Laryngectomy Management Flowchart



Appendix 3 Bedside Tracheostomy Poster

This patient has a

TRACHEOSTOMY

There is a potentially patent upper airway (Intubation may be difficult)

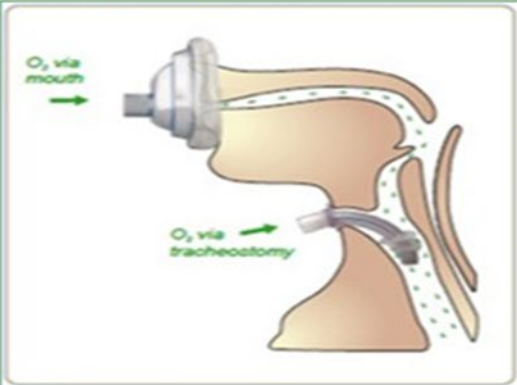
Give oxygen via BOTH tracheostomy tube/stoma and mouth

- Oxygen to face and tracheostomy tube/stoma
- Remove inner cannula/ cap
- Pass a suction catheter
- Deflate cuff
- Consider tracheostomy removal with medical guidance
- Consider ventilation via upper airway or stoma using mask or tube (*if ventilating via mouth occlude stoma site*)

Surgical / Percutaneous:

Performed on:

Tracheostomy tube size:

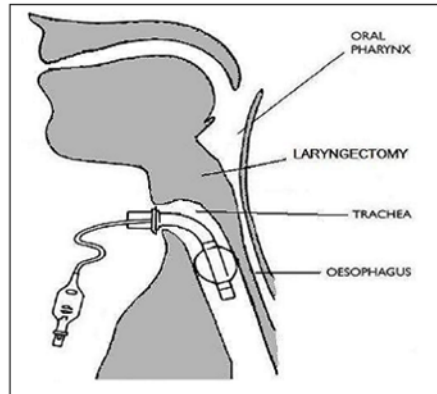


19.1.1 For an Emergency: Call 55 or Activate Code Blue

19.1.2 For advice: Call ICU 2784

This patient has a
LARYNGECTOMY
DO NOT INTUBATE, OR GIVE OXYGEN VIA THE MOUTH OR NOSE

Note: there may not be a laryngectomy tube in situ. The trachea ends at the stoma. There is no airway via the mouth or nose.



Performed on (date).....

Trachy tube size (if present)

Patient name.....

Hospital number.....

Contact numbers for help and advice at any time:

ICU: 2603 or 2784

On Call Anaesthetist: Call switchboard

ICU Physiotherapist: 2605 or page 326

Appendix 4 Tracheostomy Observation Chart

Patient NAME: Unit Number: Affix Patient Label Here	Date:	ARMADALE HEALTH SERVICE							Size of tracheostomy:		
		TRACHEOSTOMY OBSERVATION CHART									
		Single lumen tube: change every 7-14 days							Date of insertion:		
		Double lumen tube: change every 4 weeks									
		Check patient and record obs. Minimum 4 hourly							Date of last change:		
									Date of next change due:		
CARE BUNDLE ELEMENTS	TIME	TIME	TIME	TIME	TIME	TIME	TIME	TIME	TIME	TIME	Comments & keys
Check emergency bed side equipment											
Humidification checked											S = Swedish nose F&P = Fisher & Paykel
Suction Check 2 hourly (Tick number of times)											Sputum Type: C = Clear B = Blood stained P = Purulent
Inner tube cleaned (tick)											Minimum of 4 hourly
Cuff pressure (note if un-cuffed tube)											Aim for 15-25 cmH ₂ O NC = Non Cuffed CD = Cuff Deflated
Tracheostomy site and dressing checked											Check every 24hrs
Speaking Valve? (Y / N)											Time period tolerated
Oxygen concentration / flow rate											Check correct delivery
Signature:											
Discipline/ Designation											